Level-1 models at a glance

ccd_opt_ml · each model predicts an amplifier's read noise σe, dark charge λ and gain from its trace (no amplifier labels, except the reference). Last column: typical error on synthetic packets (toy CCD world), 41 packets, mean of 3 datasets, packet-grouped 5-fold cross-validation; it compares methods, not the real CCD.

Model *(looks at)*	What it does	σe/λ/gain error [%]
Baselines: no trace at all		
mean *nothing*	Predicts the average of the training labels for every example, with the training spread as its error bar. Any model worth using must beat it; the report's "vs mean" score divides every model's error by this one.	57.5/29.2/31.3
settings *settings only*	A Gaussian process on the operating settings alone: no trace and no amplifier label. It measures how much the settings explain by themselves; a trace model that cannot beat it has not shown that the traces add anything.	57.5/21.7/31.3
Standard regressors: the settings + the 85 hand-made trace numbers (own traces and the nominal reference traces)		
ridge *85 numbers*	Bayesian linear regression: a weighted sum of the inputs, with a prior that keeps the weights small, and an error bar. Simple, fast and hard to fool; a sanity check on the fancier models.	9.5/12.4/0.9
trees *85 numbers*	Gradient-boosted decision trees: a few hundred small if-then trees, each fitting what the previous ones got wrong; the error bar comes from extra fits of the 16 % and 84 % quantiles. Handles thresholds and kinks and ignores useless inputs, but cannot extrapolate.	9.6/21.0/1.3
gp *85 numbers*	Gaussian process: predicts from the training examples whose inputs are most similar, with one fitted length scale per input (how much that input matters). Smooth predictions with honest error bars, which level 2 needs; can over-fit with many inputs and few examples.	7.9/7.4/0.9
nn *85 numbers*	An ensemble of small neural networks (two hidden layers of 64); each outputs a value and an error bar, and their disagreement is added to the uncertainty. Flexible, but needs more data than the first packets will give.	9.7/16.9/1.3
Physics-based: the main models		
trace_physics *Allan curve*	No learning. Fits $\sigma(N)^2 = a^2/N + b^2$ to the trace's Allan deviation (a = noise of one skip, b = noise floor) and evaluates it at the image's N = 500 skips; divides by the average training gain. Shows how far pure physics gets.	36.1/29.2/31.3
physics (A), the default *Allan curve + settings + numbers*	Starts from trace_physics and adds Gaussian processes that learn the gain and a correction for what the trace physics misses (e.g. noise only the image sees). Training packets far off the rest are down-weighted, so one bad packet cannot bend the correction. λ comes from a SAAS GP, a Gaussian process whose prior switches off inputs the data do not need.	9.1/7.4/0.9
physics_joint *as A + λ row by row*	As A, but λ is built from the image's row-by-row dark charge: dark current and clock-induced charge (CIC) are fitted together by one two-output Gaussian process that knows their correlated errors (SAAS prior). Best λ so far.	9.1/4.6/0.9
sets (E) *each trace set*	As A, but instead of averaging the repeated trace sets first, a small network reads every set's Allan curve and learns how much to trust each (attention). Robust to damaged sets and intermittent pickup. Best σe so far.	5.7/10.9/0.9
Networks that learn their own features from the 14 trace curves		
cnn *14 curves + settings*	A 1D convolutional network that reads the curves directly (Allan deviation, spectra, autocorrelation, waveform shapes) and learns its own features; an ensemble gives the error bar. Can find effects the hand-made numbers miss, given enough data.	7.6/13.1/2.0
pretrained (D) *14 curves + settings*	The same kind of network, first pre-trained on unlabelled traces (reference traces need no image) with self-made tasks: fill in masked parts of the curves, predict the settings from the curves, and a contrastive task. Then fine-tuned on the labelled packets.	6.8/17.2/2.6
deep_kernel (C) *pre-trained summary + settings*	A Gaussian process that measures similarity on the pre-trained network's summary of the curves (kept fixed) plus the settings: learned features with Gaussian-process error bars.	8.5/20.7/0.9
Reference only		
physics_hier (A+B) *as A + told the amplifier*	Model A on hierarchical Gaussian processes told the module and amplifier: shared trends plus per-amplifier offsets. Not a candidate: comparing it with A measures what amplifier labels would add.	9.1/10.6/0.9